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MIDD Evidence to Support Drug Regulatory Decisions in China

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ABSTRACT

The paper systematically reviews the technical guidelines for Model-Informed Drug Development (MIDD) issued by China's Center for Drug Evaluation (CDE), analyzing typical application cases of innovative drugs approved both in China and internationally to evaluate MIDD's critical role in supporting drug development and regulatory decision-making. As of December 2025, CDE has released seven specific guidelines focused on quantitative pharmacology models, covering the entire drug development lifecycle, including dose exploration and optimization, pediatric extrapolation, rare disease drug development, and exposure-response (E-R) relationship research. MIDD have been successfully applied across multiple domains in innovative drug development, optimizing dosing regimens through E-R analysis for dose selection; constructing comprehensive evidence chains via modeling and simulation for pediatric and rare disease drug development; supporting dosage rationality through population PK and PBPK models for dose adjustments in special populations; and characterizing complex in vivo processes in advanced therapeutic products through mechanistic models. While MIDD enhances R&D efficiency and reduces clinical trial burdens, the field in China currently faces three major challenges: data quality, prospective design, and validation standards. It is necessary to strengthen data collection, model validation, and full lifecycle management to promote the paradigm shift from "model-assisted" to "model-driven" drug development, thereby accelerating the global market entry of Chinese innovative drugs.

Since the implementation of China's pharmaceutical review and approval system reform in 2015, with the continuous deepening of the reform, China's innovative drug development has achieved remarkable results. Over the past 5 years, the number of innovative drugs approved in China has increased annually. In 2021, a total of 47 innovative drugs were approved for market entry. By 2025, the National Medical Products Administration (NMPA) approved 76 innovative drugs, including 47 chemical drugs, 23 biologics, and 6 traditional Chinese medicines. Among the 47 chemical drugs, 38 were domestically developed, accounting for 80.85%, and 9 were imported. Of the 23 biologics, 21 were domestically developed (91.30%) and 2 were imported [1]. China's pharmaceutical industry is gradually shifting from a market dominated by generics to one driven by innovation, with an increasing number of drugs developed in parallel with global

standards. Particularly since China joined the International Council for Harmonization (ICH), technical standards have aligned with international benchmarks. The adoption of ICH guidelines has facilitated the global development of domestically produced innovative drugs. Additionally, by adhering to principles of scientific rigor and efficiency, the NMPA has issued a series of technical guidelines that further support the acceleration of China's pharmaceutical industry toward an innovation-driven model.

Model-Informed Drug Development (MIDD) leverages modeling and simulation technologies to integrate physiological, pharmacological, and disease process information for quantitative data analysis. This approach deepens the understanding of drug mechanisms, characteristics, and the role of drug intervention

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in disease progression. MIDD supports the evaluation of benefit–risk ratios, the rationality of dose selection, the adjustment of dosing regimens in subpopulations, and the preparation of drug labeling, thereby facilitating drug development and supporting regulatory decision-making. In 2020, the Center for Drug Evaluation (CDE) issued the *Guidance on Model-Informed Drug Development*, marking China's formal entry into a new era of MIDD. Subsequently, the CDE has issued multiple technical guidelines related to quantitative modeling to address key technical challenges, including dose selection and optimization, pediatric extrapolation, and the optimization of clinical development strategies for rare diseases.

Along with MIDD frameworks of the Food and Drug Administration (FDA), and the European Medicines Agency (EMA), China CDE, FDA and EMA follow the principles of ICH M15, centering on the Context of Use (CoU) and adopting a risk-based approach to model validation. China issued independent guidelines early and rapidly established a domestic system, balancing scientific rigor with industrial accessibility; MIDD is widely adopted among innovative drugs, though early interaction mechanisms remain evolving, such as MIDD prospectively influencing trial design.

This paper systematically reviews the series of MIDD-related technical guidelines issued by the CDE, along with typical case studies of innovative drugs approved in China and globally over the past 5 years. Furthermore, it analyzes and elucidates the critical role of MIDD in supporting drug development and regulatory decision-making in China. In these cases, the applied models—including PPK, PBPK, PKPD, and E-R models—are well developed and rigorously validated; most importantly, all models comply with the corresponding regulatory guidelines.

1 | MIDD-Related Technical Guidelines Issued in China

On December 31, 2020, the *Guidance on Model-Informed Drug Development* was released, marking the first MIDD technical guideline globally. Currently, ICH is actively coordinating the development of *M15: General Principles for Model-Informed Drug Development* to harmonize these standards globally.

As of December 2025, the CDE has issued seven technical guidelines specifically targeting quantitative pharmacology models. Additionally, several other guidelines, such as the *Guidance on Clinical Pharmacology Research for Innovative Drugs* and the *Guidance on Clinical Pharmacology Research for Cell Therapy Products*, have integrated modeling and simulation to fill data gaps. These applications provide critical decision-making and regulatory support for innovative drug development, including advanced modalities like cell therapy products (Table 1).

2 | Application of MIDD-Related Technologies in China

Driven by clinical needs, innovative therapeutic agents for severe, chronic, and rare diseases have been widely approved, offering patients more comprehensive treatment options. As

representative treatments, Cell and Gene Therapy (CGT) products have become critical therapeutic pathways for refractory and rare diseases. The market entry of multiple domestically developed CGT products marks significant progress in this cutting-edge field.

To support drug development and regulatory decision-making, various modeling approaches, such as PopPK, PBPK, PK/PD, and quantitative systems pharmacology (QSP) models, are increasingly employed. This paper illustrates the efficient application of MIDD in China through case studies of approved innovative drugs. These cases demonstrate how MIDD has provided robust scientific evidence to resolve challenges in dose selection, benefit–risk assessment, special population dose adjustment, pediatric dose extrapolation, and drug interaction evaluation, thereby accelerating product development and market entry (Figure 1).

2.1 | Comprehensive E-R Analysis via MIDD Provides Multidimensional Information for Dose Selection

Dose exploration and optimization are critical components of innovative drug development. By integrating drug and disease characteristics, model-informed dose exploration employs mathematical models and statistical methods to quantify data across multiple research stages. This approach facilitates a comprehensive analysis of drug exposure, safety and efficacy, providing scientific evidence for dose selection and supplementary evidence for clinical evaluations, thereby reducing trial uncertainty. Throughout the development process, the continuous integration of non-clinical data with clinical PK/PD information allows for robust E-R analysis. Ultimately, selecting the appropriate exposure metrics, efficacy indicators, and modeling methodologies is foundational to determining optimal dosing regimens.

Approved in China in 2023, Tafolecimab Injection is the first domestically developed and third globally approved PCSK9 monoclonal antibody inhibitor [9]. As cardiovascular disease rates continue to climb, Tafolecimab represents a significant advancement in lipid-lowering therapy for Chinese patients and holds potential for broader global application. Its subcutaneous dosing regimen of every 6 weeks significantly improves patient adherence, providing a more accessible treatment option. During the development, PopPK/LDL-C models were established using data from six clinical trials conducted in China. While key covariates such as body weight, gender, baseline LDL-C, and disease status were found to affect model parameters, none necessitated dose adjustments. E-R analysis indicated that a steady-state plasma trough concentration (C_{trough}) of 5 µg/mL achieved the maximal reduction in LDL-C, with no further efficacy gains observed at higher concentrations [10]. The absence of exposure-dependent safety signals supports the optimized 6 week dosing frequency.

Ganagliflozin Proline is a sodium-glucose cotransporter 2 (SGLT2) inhibitor that competitively binds to the glucose-binding site of the transporter, blocking SGLT2-mediated renal glucose reabsorption and lowering blood glucose by increasing urinary glucose excretion (UGE). This drug was approved for market entry in China in January 2024 for the treatment of

TABLE 1 | Summary of MIDD-related technical guidelines issued in China.

Title and issue date	Applicable scenario	Key highlights
Guidance on Model-Informed Drug Development [2], December 31, 2020	Applicable throughout the drug development lifecycle, spanning preclinical research, all phases of clinical trials, and post-marketing surveillance, supporting dose selection, population optimization, protocol design, and regulatory decision-making, enhancing scientific rigor and development efficiency.	The first MIDD guidance principle globally, introducing modeling and simulation methods to support decision-making across the entire drug development lifecycle, improving research efficiency and promoting alignment with international standards.
Guidance on Population Pharmacokinetic Studies [3], December 31, 2020	Clarifying the process and quality control requirements for population PK (PopPK) studies, standardizing data management, model construction, and validation, emphasizing covariate analysis, guiding sparse sampling design, and supporting individualized dosing and R&D efficiency.	This guideline systematically standardizes the core elements of PopPK studies. In particular, it strengthens the scientific requirements for study design, model development, validation, and data compliance, while clarifying the evidentiary value of PopPK in key regulatory decisions, such as dose optimization for special populations.
Guidance on the Application of Physiologically Based Pharmacokinetic Models in Pediatric Drug Development [4], March 28, 2023	Applicable to pediatric new drug development by utilizing physiologically based pharmacokinetic (PBPK) models to simulate and predict pediatric pharmacokinetic characteristics, supporting extrapolation of adult data to pediatric populations, optimizing clinical trial design, and reducing the need for unnecessary pediatric trials.	This guideline emphasizes the critical role of PBPK models in pediatric drug development, aiming to optimize pediatric clinical trial design through modeling and simulation, provide scientific support for dose selection and data extrapolation, and encourage early communication with regulatory agencies regarding these strategies.
Guidance on Quantitative Methodology for Extrapolating Adult Drug Data to Pediatric Populations [5], April 12, 2023	Applicable to pediatric drug development by leveraging existing adult data through quantitative methodologies (e.g., PopPK, PBPK, PK/PD models) to extrapolate adult data to pediatric populations, thereby optimizing R&D strategies.	The first domestic guidance on quantitative methodologies for pediatric extrapolation, standardizing model-driven methods, providing statistical and modeling frameworks, improving pediatric drug R&D efficiency, and promoting accessibility of pediatric medications.
Guidance on Model-Informed Dose Exploration and Optimization for Innovative Drugs [6], December 19, 2024	Applicable during the clinical research phase of innovative drugs, using modeling and simulation to integrate multidimensional data for exploring and optimizing dosing regimens (dose, interval, duration), supporting dose selection and trial design.	Standardizes model-informed dose exploration methods, reducing unnecessary clinical trials, improving R&D efficiency and success rates, and providing a rigorous scientific basis for confirmatory clinical trials.

(Continues)

TABLE 1 | (Continued)

Title and issue date	Applicable scenario	Key highlights
Guidance on Model-Informed Development of Drugs for Rare Diseases [7], June 15, 2025	Focusing on addressing core challenges in rare disease drug development, such as patient scarcity and ethical dilemmas, systematically explaining how quantitative models support key decisions across the entire clinical development lifecycle, from strategy formulation to post-marketing studies.	The release of this guidance marks China's rare disease drug development entering a new era of 'model-driven evidence integration'.
Guidance on Exposure-Response Relationship Research [8], July 30, 2025	Covers the entire drug development lifecycle, systematically guiding how to convert non-clinical data, select doses, adjust for special populations, and manage post-marketing changes through quantitative E-R relationships, supporting key R&D and regulatory decisions.	First to establish a comprehensive standardized framework covering data, models, and reporting, elevating E-R relationships to a core evidence standard. Emphasizes model-based analysis and dynamic iteration, providing a scientific decision-making pathway for clinical endpoint selection and dose optimization.

patients with type 2 diabetes mellitus (T2DM). Studies on marketed SGLT2 inhibitors have confirmed that UGE and its change from baseline (ΔUGE) can serve as pharmacodynamic (PD) indicators for this class of drugs. Phase I clinical trial results for ganagliflozin showed that within a dose range of 10 mg–50 mg, the 24 h change in UGE (ΔUGE_{0-24}) increased dose-dependently, with the 50 mg and 100 mg doses showing similar ΔUGE_{0-24} values. A PK/PD model was established using ΔUGE_{0-24} as the PD indicator. Dose-exposure-response analysis indicated that a 25 mg dose achieved half-maximal effective concentration (EC_{50}). Based on the simulated steady-state corrected change in UGE ($\Delta\text{UGE}_{\text{c}}$, ss) for Chinese T2DM patients, the established relationship between $\Delta\text{UGE}_{\text{c}}$, fasting plasma glucose (FPG), and glycated hemoglobin (HbA1c) for SGLT2 inhibitors was utilized to further simulate HbA1c curves at weeks 1, 4, 8, and 24. It was projected that after 24 weeks of administration, the HbA1c reduction from baseline for the 25 mg and 50 mg dose groups would achieve the efficacy endpoint targets of similar drugs in this class. Consequently, the 25 mg and 50 mg doses were recommended for Phase III trials, a decision subsequently validated by the trial results [11].

Mechanistic models typically integrate disease pathophysiology and drug mechanisms, incorporating extensive multidimensional data to provide an interpretable, quantitative assessment of drug PK, efficacy, and safety. QSP model integrates the relationships between diseases, drugs, and biological systems, making it particularly effective for dose section in complex, innovative therapies such as bispecific antibodies. The China NMPA approved Teclistamab, subcutaneous, bispecific antibody targeting BCMA and CD3 in June 2024. It is indicated for adults with relapsed or refractory multiple myeloma (RRMM) who have received at least three prior therapies, with a recommended dose of 1.5 mg/kg once weekly (QW). Using PopPK models, E-R analysis, and QSP modeling, the feasibility of switching to 1.5 mg/kg every 2 weeks (Q2W) for patients maintaining a complete response (CR) or better lasting ≥ 6 months was systematically evaluated. Based on data from the MajesTEC-1 study, modeling and simulation showed that the median serum trough concentrations (C_{trough}) after the first and fourth Q2W doses were 14.4 $\mu\text{g/mL}$ and 11.7 $\mu\text{g/mL}$, respectively. While these levels are lower than the QW C_{trough} of 20.4 $\mu\text{g/mL}$, they remain above 90% of the maximum effective concentration. The model simulated virtual patient populations to quantitatively describe RRMM patient characteristics and treatment responses, capturing both population averages and individual variability. Results indicated that, despite the lower C_{trough} , the Q2W regimen yielded response rates comparable to QW in terms of target cell-drug-effector cell (TBE) trimer formation, tumor volume reduction, and duration of response (DOR) [12].

2.2 | MIDD Forms a Comprehensive Evidence Chain for Rare Disease Treatments and Pediatric Populations

Quantitative pharmacology plays a critical role in rare disease drug development by synthesizing all available data to uncover the intrinsic patterns of disease progression and drug action. By maximizing the utility of limited data, these studies enable a robust assessment of clinical benefits and risks. This

Application of MIDD-Related Technologies in China

Dose Selection via Exposure-Response Analysis

Drug	Indication	Model	Key Finding
Tafolecimab (2023)	Hypercholesterolemia	PopPK / LDL-C	C _{trough} >= 5ug/mL, 6-week dosing interval
Ganagliflozin (2024)	Type 2 Diabetes	PK/PD	25mg reaches EC ₅₀ , Phase III recommended
Teclistamab (2024)	Multiple Myeloma	PopPK + QSP	Q2W and QW efficacy comparable, supports optimization

PopPK/PKPD/QSP models support dose selection and optimize dosing regimens

Evidence Chain for Rare Diseases & Pediatrics

Drug	Indication	Method	Value
Ritlecitinib (2023)	Alopecia Areata (incl. adolescents)	PopPK + E-R	First China-approved for adolescents in this indication
Stapokibart (2024)	Atopic Dermatitis (pediatric)	Exposure-matching + Modeling	First domestic IL-4R α antibody; fills therapeutic gap of pediatric patients

Model-driven evidence integration · Quantitative pharmacology · Pediatric extrapolation strategy

Quantitative Evidence for Special Population Dosing

Drug	Indication	Model	Key Finding
Netasavir/Encofovir (2025)	Hepatitis C	PopPK	No dose adjustment needed for mild renal impairment/elderly
Dorzagliatin (2022)	Type 2 Diabetes	PBPK	Minimal food effect; clinical DDI study waiver supported

Characterizing in-vivo behavior of Complex Mechanism drugs

Drug	Representative	Method	Value
CAR-T Cell Drug	7 products approved domestically	Mechanistic PK/PD	CAR-T expansion, CRS toxicity prediction
siRNA Gene Therapy	RBD1016 (HBV)	Semi-mechanistic PK/PD	Predict hepatic concentration, HBsAg level changes

MIDD successfully applied throughout innovative drug lifecycle | Data source: Center for Drug Evaluation, NMPA | As of December 2025

FIGURE 1 | The application of MIDD-related technologies in China.

evidence-based approach informs development and regulatory decisions based on favorable benefit-risk ratios, effectively reducing uncertainty and increasing the overall success rates in rare disease drug development.

Ritlecitinib Tosylate inhibits JAK3-dependent cytokine signaling and TEC tyrosine kinase family-dependent immune receptors involved in the pathophysiology of alopecia areata. Approved in 2023 for the treatment of severe alopecia areata in adolescents (12 years and older) and adults, this drug is the first in China to be approved for adolescent patients, meeting an urgent clinical need. Alopecia areata is a specific autoimmune disease with a rising prevalence, particularly among younger populations. Long-term hair loss significantly impacts patients' psychological health and quality of life, especially adolescents. Clinical data showed that a once-daily 50 mg dosing regimen achieved the primary endpoint of $\geq 20\%$ reduction in the Severity of Alopecia Tool (SALT) score at Week 24. Results from the Chinese subgroup were consistent with the overall population. Population PK analysis revealed no clinically relevant differences in PK across age (12–73 years), weight, gender, or race [13]. E-R analysis used the average concentration (C_{avg}) during the time interval between the previous SALT score and the current SALT score as the PK exposure metric driving efficacy. The estimated EC_{50} was 53.6 ng/mL (C_{avg}), indicating that the highest maintenance dose of 50 mg was close to the half-maximal effective dose (ED₅₀) (the C_{avg} for 30 mg and 50 mg once daily were 27.0 and 52.0 ng/mL, respectively). Poisson regression was used to describe the relationship between C_{avg} exposure and safety events, such as infections, herpes zoster, and rashes. Results showed that the incidence of infections and rashes increased with concentration, with a slight increase observed when the

dose was escalated from 30 mg to 50 mg. Based on population PK analysis and E-R relationships, weight, gender, and disease severity were identified as key covariates affecting exposure. Combined with safety data from Chinese patients, the potential risks were comprehensively evaluated, leading to the recommendation to include key covariates affecting exposure in the drug label and to further accumulate safety information post-marketing.

Dose extrapolation is a common strategy in pediatric drug development, founded on the premise that pediatric and adult populations share sufficient similarities in disease pathophysiology, drug pharmacology, and expected therapeutic response. Because inappropriate dose selection is the primary cause of failure in pediatric drug development programs, utilizing scientific and robust quantitative methods is essential to address these challenges. Stapokibart Injection, approved in China in September 2024, is the first domestically approved and the second globally approved IL-4R α antibody. It fills a critical gap in biologic treatment options for atopic dermatitis (AD) in China. While currently approved for AD, chronic sinusitis with nasal polyps, and seasonal allergic rhinitis in adults, there is an urgent need for pediatric applications. Pediatric AD is characterized by early onset, high incidence, frequent recurrence, and prolonged duration. Given the similarities between adult and pediatric populations regarding disease risk factors, skin inflammation, and treatment response, researchers initiated clinical studies for pediatric AD. During development, the principle of exposure matching was employed, using modeling and simulation to determine pediatric dosing. Following Phase Ib/IIa studies in children aged 6–11 years, preliminary safety, efficacy, and PK data were established for this age group. Optimal dosing was ultimately

selected by matching weight-based exposure levels to those achieved by the approved adult dose [14].

2.3 | MIDD Provide Quantitative Evidence for Dose Adjustments Under Different Usage Scenarios

Dose selection is typically based on general patient populations enrolled in clinical trials. However, factors such as advanced age and varying degrees of organ impairment can significantly alter drug pharmacokinetics, necessitating potential dose adjustments to maintain efficacy and safety. While conducting independent PK studies in these specific populations remains the gold standard, it is not always feasible due to safety margins, complex metabolic profiles, mechanism of action, or practical development constraints. In such cases, PopPK or PBPK models are utilized to characterize drug exposure. When integrated with established E-R relationships, these models could provide a robust scientific basis for justifying dose adjustments in special populations, in some situations, without the need for additional standalone clinical trials.

Netanasvir Phosphate Capsule/Encofosbuvir tablet combination is a domestically developed innovative anti-HCV therapy launched in China in February 2025. It represents China's first fully domestically owned, oral pan-genotypic HCV treatment regimen. This approval significantly advances the public health goal of “eliminating viral hepatitis by 2030” established by the World Health Organization (WHO). Population PK analysis of the Netanasvir Phosphate clinical program showed netanasvir exposure was comparable between patients with mild renal impairment (creatinine clearance 56.6–225.2 mL/min) and those with normal renal function. Furthermore, age (20–77 years) and gender were found to have no clinically significant impact on netanasvir exposure. Consequently, the approved dosing regimen covers elderly patients (≥ 65 years) and those with mild renal impairment without the need for dose adjustments.

Dorzagliatin Tablet is the first-in-class glucose kinase activator (GKA) to be approved globally, receiving marketing authorization in 2022 for the treatment of type 2 diabetes. Using a validated PBPK model, researchers simulated the impact of high-fat, high-calorie dietary conditions (e.g., prolonged gastric transit time) on Dorzagliatin exposure. The simulation indicated that even extreme dietary variations in patients with type 2 diabetes would not significantly alter drug exposure, a finding consistent with clinical PK results in Chinese patients that compared dosing 1 h versus 4 h post-meal [15]. Furthermore, the model simulations were conducted to predict exposure under various drug–drug interaction (DDI) scenarios involving CYP3A4 enzymes. Results showed that co-administration with moderate CYP3A4 inhibitors increased Dorzagliatin $C_{\max}$ and AUC to 1.20- and 2.40-fold, respectively. In contrast, weak CYP3A4 inhibitors increased $C_{\max}$ and AUC by no more than 8% and 20%, which are not considered clinically significant based on established E-R relationships. Conversely, moderate CYP3A4 inducers decreased $C_{\max}$ by 23% and AUC by 51% [16]. Consequently,

these PBPK findings provide a robust scientific basis to support the exemption from standalone clinical DDI studies with moderate and weak CYP3A4 inhibitors and inducers.

2.4 | MIDD Characterize the in Vivo Behavior of Complex Mechanism Advanced Therapeutic Products

Cell and gene therapy (CGT) has emerged as the third industrial revolution in global biopharmaceutical innovation. China has made remarkable progress in this field with nine CGT drugs approved for market entry, including seven CAR-T therapies, one conditionally approved stem cell therapy, and one fully approved gene therapy. These milestones demonstrate China's capacity to both contribute to and compete within the global biopharmaceutical ecosystem [17]. CAR-T cell therapy uses genetic engineering to modify T cells to express chimeric antigen receptors (CARs), targeting specific antigens such as CD19 and BCMA to treat tumors and systemic lupus erythematosus.

Upon administration, CAR-T products exert therapeutic effects via complex biological signaling pathways and sustained proliferation; their cell kinetics reflect the dynamic interactions between CAR-T cells and the host. The in vivo behavior is shaped by physiological status, treatment interventions, and product-specific characteristics. Due to these multiple confounding factors, it is difficult to determine a product's in vivo trends using empirical models alone. Mechanistic models integrating physiological, immunological, CAR-T cell phenotypes, and cytokine-related information are required to explore the relationships between cell subsets, secreted cytokines/chemokines, and disease states. Such models guide clinical dosing design, describe correlations between cell subsets, expansion, activation characteristics, and clinical efficacy indicators to optimize product phenotypes, and characterize cell distribution and cytokine release to analyze potential toxicities such as Cytokine Release Syndrome (CRS), hematological and neurological toxicities [18].

Gene therapy efficacy depends on the concentration at the target site within the cytoplasm; high tissue distribution and accumulation often result in a long pharmacodynamic half-life and sustained efficacy. However, because it is difficult to directly detect long-term drug concentrations at human target sites, establishing human PK-PD correlations is challenging. For example, oligonucleotides are rapidly degraded by endogenous nucleases, resulting in a short plasma half-life where systemic exposure and clinical efficacy often diverge. Advanced delivery systems can alter the tissue distribution of gene therapy products. For GalNAc-conjugated oligonucleotides, PBPK/PD models can dynamically predict blood-to-liver concentrations. Typically, animal PK/PD models are established and extrapolated across species, followed by human models that use clinical data for calibration to predict long-term liver concentrations and efficacy dynamics. RBD1016, an siRNA drug for chronic hepatitis B (HBV), utilizes a structure that binds to N-acetylglucosamine to promote liver delivery. A semi-mechanistic model was constructed for RBD1016 to elucidate PK/PD characteristics in mice and PK characteristics in monkeys, laying the foundation for clinical translation [19]. This model accurately fit the PK and PD

data in HBV-transgenic mice and reasonably described monkey plasma and liver concentrations. Simulation results indicated the model could predict HBsAg levels after multiple dosing, supporting the effective exposure range for first-in-human trials. Clinical data for RBD1016 will further refine the model to support subsequent development stages.

3 | Challenges and Future Perspectives

MIDD is rapidly becoming a core driver in the pharmaceutical industry. It represents more than just a technical tool; it is a paradigm shift transitioning from traditional “linear, trial-and-error” models to “cyclical, predictive” models. However, models inherently carry uncertainty; therefore, rigorous modeling and thorough validation are required to ensure scientific validity and reliability. Throughout all stages of model development, good modeling practices should be adopted to enhance model credibility, reduce uncertainty, and optimize predictive capability.

Taking a comprehensive look at domestic development and regulatory practices, MIDD in China is currently facing three major challenges: data quality, prospective design, and validation standards. Scientifically reliable models depend on high-quality data; however, insufficient or incomplete data collection constrains the reliability of model conclusions. Furthermore, multiple issued guidelines emphasize prospective data collection, explicitly requiring that model analysis be planned in parallel with study protocol development. This involves the systematic collection of demographics, physiology/pathology, disease progression, exposure, efficacy, and safety data to form a “learn-confirm” dynamic closed loop. At the same time, model validation must be matched with regulatory decision-making risks. Critical models must undergo external dataset validation, sensitivity analysis, and uncertainty assessment, establishing a full lifecycle management mechanism. Only through the synergy of these three aspects can the transition from “model-assisted” to “model-driven” be realized.

While MIDD is widely applied in small-molecule and biologic development, there are notable opportunities to expand its use in advanced therapeutic products (e.g., cell/gene therapies, nucleic acid drugs) and locally acting products (e.g., topical, ocular, and inhalation formulations). The mechanisms of advanced therapeutic are highly complex, and traditional PK/PD models often fail to capture these processes, leading to prediction bias by omitting factors like cell heterogeneity and the tissue microenvironment. There is an urgent need for integrated mechanistic models that incorporate multi-omics data, cell kinetics, and tissue microenvironment to simulate in vivo trajectories and gene editing efficiency. For locally acting products, efficacy depends on precise delivery to the target site, influenced by the site's physiological structure (e.g., skin stratum corneum barrier, blood-eye barrier), formulation release characteristics, and local metabolic pathways. Because existing mechanistic models often rely on systemic exposure rather than on local concentration-effect relationships, making it difficult to guide formulation optimization. Therefore, developing physiologically based local pharmacokinetic (L-PBPK) models is essential. These models integrate formulation release kinetics with target site physiology to quantitatively describe drug behavior in local tissues.

4 | Conclusions

China's innovative drug development has transitioned from the extensive development model driven by quantity and fast follow to a new high-quality development stage focused on first-in-class innovation. Adopting advanced methodologies and cutting-edge tools aims to raise the success rate of new drug research and development. By applying scientifically sound models to support drug development progress and regulatory decision-making, more innovative drugs can be launched sooner to benefit patients worldwide.

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Conflicts of Interest

The authors declare no conflicts of interest.

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